Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/30/2023
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	High Dimensional Biology to Understand the Functional Response of Salmonella to Long-Term Multigenerational Growth in the Chronic Stress of Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	12/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	Ground-based spaceflight analogue NOTE: Extended to 12/31/2022 per F. Hernandez/ARC (Ed., 7/27/21) NOTE: Extended to 12/31/2021 per NSSC information (Ed., 3/8/21) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 7/20/2020) NOTE: Extended to 4/30/2020 per NSSC information (Ed., 6/26/18) NOTE: Extended to 5/31/2019 per F. Hernandez/ARC (Ed., 5/8/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Abu-Ali, Galeb D.V.M., Ph.D. (Harvard College School of Public Health) Barrila, Jennifer Ph.D. (Arizona State University) Huttenhower, Curtis Ph.D. (Harvard College School of Public Health) Ott, Mark Ph.D. (NASA Johnson Space Center) Travisano, Michael Ph.D. (University of Minnesota)		
Grant/Contract No.:	NNX15AL06G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	While unexpected responses in microorganisms are well documented for short duration culture, the impact of long duration, multigenerational growth of microorganisms in the chronic stress of microgravity has not been investigated. The implications of this gap in knowledge are tremendous, as the response to short duration growth in microgravity alters gene expression, pathogenesis-related stress responses (ex., biofilm formation and resistance to antibiotics), and virulence. Moreover, it is known that stress responses (including chronic stress) can increase mutagenesis in bacteria leading to genomic plasticity, which may provide novel mechanistic insight into genomic instability underlying resistance to antibiotics, pathogenicity, and virulence. We propose that long-term growth in spaceflight culture will induce heritable genomic and epigenetic changes in microorganisms with adverse phenotypic outcomes that could have a profound impact on the risk to crew health and vehicle system performance. Our previous studies aboard the Space Shuttle investigated the response of the model enteric pathogen, Salmonella enterica serovar Typhimurium (S. Typhimurium) to short duration spaceflight culture and demonstrated a significant increase in virulence in a murine model of infection, in addition to biofilm formation, as compared to identically grown cultures on the ground. In addition, global microarray and proteomic analyses revealed that 167 transcripts (including small noncoding RNAs/ncRNAs) and 73 proteins changed expression, with the evolutionarily conserved protein Hfq identified as a likely global regulator involved in the response to this environment. Hfq binds to small ncRNAs thereby facilitating their association with mRNAs, the outcome of which plays a diverse role in regulating gene expression, virulence, stress responses, antibiotic resistance, DNA methylation, and genome stability. We hypothesize that long-term multigenerational culture of S. Typhimurium in the spaceflight environment will result in stress-induced mutations and genomic instability as reflected in changes in the genomic, epigenetic, transcriptomic, and select phenotypic characteristics (including virulence) that will reflect a central role for Hfq as a global regulator of these heritable responses. To address this hypothesis, we propose to characterize the impact of long term spaceflight culture (at least 100 generations) of an S. Typhimurium wild type and hfq mutant on the genomic, epigenetic, transcriptomic, virulence, and pathogenesis-related characteristics of these strains. This proposal addresses Research Emphasis 2 - ISS (International Space Station) Microbial Observatory Studies from section B.3 of this announcement, as it is designed to “take advantage of the technological maturity, low cost, and speed of genomic analyses and the rapid generation time of microbes to monitor the evolution of microbial genomic changes in response to the selective pressures present in the spaceflight environment.” In addition, by employing genomic, transcriptomic, and epigenomic analysis, this proposal also addresses the Research Emphasis 1 – Spaceflight Omics Studies in section B.2. In its design, this proposal will address the NASA Space Biology Plan, Overarching Questions CMM-1a, CMM-1d, and CMM5. This proposal also directly addresses the highest priority recommendations from the Decadal Survey for the establishment of an ISS Microbial Observatory conducting long-term, multigenerational studies of microbial population dynamics (Priority P1). In addition, this study analyzes “microbial growth and physiological responses to the multiple stimuli encountered in spaceflight environments,” as described by the panel in Priority P2.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	Research Progress: Completed ground-based spaceflight analogue culture in the Rotating Wall Vessel (RWV) bioreactor in triplicate for both wild type and mutant Salmonella strains for 100 days. We are in the process of completing all assays (-omics profiling and phenotypic profiling). Results will be included in a manuscript to be submitted in the future. Invited Presentations: J. Barrila, S. Gangaraju, H.A. Lorenzi, H. Bean, C.M. Ott, C.A. Nickerson. Contributions of the microbiome in astronaut health: a new dimension in modeling crew infectious disease risks. 44th Committee on Space Research (COSPAR) Scientific Assembly, Athens, Greece, July 16-24, 2022. *(Note: Jennifer Barrila was the invited speaker for this seminar to present our team’s research). Invited Keynote Speaker, American Society for Microbiology (ASM) South Central Branch Meeting, Shreveport, Louisiana, October 27-29, 2022. Invited Speaker, Uniformed Services University of the Health Sciences. Bethesda, MD, November 7, 2022. Invited Speaker, Role of RpoS in regulating stationary phase Salmonella Typhimurium pathogenesis-related stress responses under spaceflight analogue low fluid shear force conditions. American Society for Space and Gravitational Research (ASGSR) Meeting, Houston, TX, Nov 2022.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Barrila J, Yang J, Franco K, Yang S, Davis T, Aronow BJ, Bean H, Davis RR, Forsyth RJ, Ott CM, Gangaraju S, Kang B, Hanratty B, Nydam SD, Kong W, Steel J, and Nickerson CA. "Spaceflight analogue culture enhances the host-pathogen interaction between Salmonella and a 3-D biomimetic intestinal co-culture model. " Front Cell Infect Microbiol. 2022 May 31;12:705647. https://doi.org/10.3389/fcimb.2022.705647 , May-2022